UNIVERSITY of SOUTH FLORIDA

# Machine Learning: Student Proud Project

The **Student Proud Project** is a semester-long, independent project where you will plan and execute a machine learning project from start to finish.

**Goal:** By the end of the semester, you will create something that makes you feel genuinely proud of yourself. It can be a working model or an insightful analysis of machine learning. This project is meant to help you build a portfolio piece that you can **showcase publicly** (e.g., GitHub, personal website), **list on your resume for career development.**

The project is divided into **two project milestones,** discussed below.

## Milestone I: Literature Survey, Reproduction of an Existing Work, and Project Proposal

**Due Date:** October 23, 2026. 11:59PM
**Deliverable:** A ~5-page document
**Percent of Final Grade:** 10%

1. **Find your interest.**
2. **Project Title:** One-line title that clearly summarizes your project idea.
3. **Introduction:** Explain your motivation, including what problem you are tackling and why does it matter?
4. **Existing Works:** Summarize at least **five recent research papers** or applied projects related to your topic.
    a. For each, briefly describe the main idea, methods, and results.
    b. Highlight what you find inspiring or useful.
5. **References:** Provide full citations.
6. **Reproduce one existing research paper**
    a. Implement the model/method from scratch (you may use existing ML libraries like PyTorch, TensorFlow, Numpy, Pandas, or scikit-learn).
    b. Run the experiment on the **same dataset** (or a close equivalent if unavailable).
    c. Compare your results to the original results and explain any differences.

## Milestone II: Novel Contributions

**Due Date:** November 25, 2026. 11:59PM
**Deliverable:** Final code and final report (~5 pages, excluding previous milestone)
**Percent of Final Grade:** 15%

In this milestone, you will move beyond reproduction and add your own contribution.

What counts as a contribution? Examples include:

- Add or modify features for improving results and discuss educated rationale
- Changing model architecture and discuss why
- Combining methods from multiple papers
- Analyzing limitations

Your final report should include:

1. Methodology (explain your contributions).
2. Experiments (datasets, settings, metrics).
3. Results (tables, figures, comparisons).
4. Discussion (what worked, what did not, what you learned).

**Tip:** Even small improvements or creative experiments count as contributions. The goal is to show independent thinking.

## Important Materials

- **Where to find papers:**
    - Google Scholar
    - ArXiv.org
    - Ask Generative AI (copilot, chatgpt, so on)
- **How to read a paper effectively:**
    - Start with the abstract, introduction, and conclusion to get the big picture
    - Look at the figures, tables, and experiments before diving into technical details
    - Focus on the **problem, method, and results-**you do not need to understand everything
    - Make notes about what inspires you or what you might improve